Curriculum Vitæ

Juliette Grosset

— October 21, 2026 —

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October 21, 2026

Personal Details

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Executive Summary

Current Position:
- Post-Doctoral Researcher, IRL CROSSING, CNRS, Naval Group (Australia)
- **Qualification CNU 27** (Computer Science)

Fields: Robotics, Distributed Systems, Artificial Intelligence, Simulation, Cooperative Autonomy, Maritime Context, ITS Context, Human-in-the-loop

Awards:
- **Best Thesis Award – SPIN Doctoral School (2025)** – jury of 13 members
- Best Paper Award – ASPAI 2024
- Best Presentation Award ex-aequo – Colloque IMT 2024
- Best Poster Award ex-aequo – Colloque IMT 2022
- 4th Jury Prize, Ma Thèse en 180 secondes, Rennes qualifying round 2024

Publications:
- **Published (22)**
  - International journals: 6 (**3 Q1**)
  - International conferences: 8 (**1 rank A**: IROS)
  - 1 book chapter, 3 national communications, 4 posters
- **Under review (9)**
  - International journals: **3 Q1**
  - International conferences: 6 (**4 ICRA**: rank A*, **1 AAMAS**: rank A, 1 OMAE)
- h-index: 5 – 137 citations

Links: Google Scholar profile github.com/grossetjuliette

Supervision: 1 PhD (co-supervision), 1 Research Engineer, 13 Interns, 3 Project groups

Mobilities:
- Adelaide, Australia (2 years, 2025–2026) - Post-Doctoral Researcher
- Singapore (1 week, 2025) - Post-Doctoral Researcher
- Tokyo, Japan (3 months, 2023) - mobility during my PhD
- Bangkok, Thailand (3 months, 2019) - research internship
- Sherbrooke, Canada (4 months, 2018) - academic semester

Positions and Qualifications

Positions

2025 – present: Post-Doctoral Researcher, Naval Group / IRL CROSSING, Adelaide (Australia).
2024 (3 months): Research Engineer, IMT Atlantique / IRISA, Rennes.
2021 – 2024: PhD Student, ECAM Rennes - Louis de Broglie / IMT Atlantique / IRISA, Rennes.
2021 (1 month): Laboratory Technician, IMT Atlantique LabSticc, Brest.
2021 (6 months): Research Intern, IMT Atlantique / LabSticc, Brest.
2021 (1 semester): Pedagogical tutor, ENIB, Brest.
2020 (5 months): Assistant engineer intern, INRIA / IRISA, Rennes.
2019 (3 months): Technical intern, King Mongkut's University of Technology (KMUTT), Bangkok (Thailand).
2018 (4 months): Academic exchange, Université de Sherbrooke (Canada).

Education

2024: PhD Thesis, Computer Science, IMT Atlantique.

Institution	IMT Atlantique, Rennes, France
Title	Collective intelligence strategies for efficient autonomous industrial vehicles
PhD Supervisors	Jean-Marie Bonnin (Professor at IMT Atlantique) Alain-Jérôme Fougères (Professor at ECAM Louis de Broglie)
Reviewers	Leila Merghem-Boulahia (Professor) Gérard Chalhoub (Professor)
Jury Members	Maroua Bouzid (Professor, University of Caen Normandy, *Chair*) Leila Merghem-Boulahia (Professor, University of Technology of Troyes) Gérard Chalhoub (Professor, University Clermont Auvergne) Cédric Buche (Professor, ENIB, Naval Group Pacific - Adelaide, Australia)
Supervision	Moïse Djoko-Kouam (Lecturer-Researcher at ECAM Louis de Broglie)

2021: Master's degree SIIA (Systèmes Intelligents, Interactifs et Autonomes), UBO / ENSTA Bretagne / IMT Atlantique / ENIB, Brest, France.
2021: Diplôme d'Ingénieur (Engineering degree), ENIB, Brest, France.
2018: Academic exchange semester, Université de Sherbrooke, Canada.

Career Overview

I obtained a double degree in 2021: an Engineering diploma from ENIB (Brest) and a Master's degree in Intelligent, Interactive and Autonomous Systems (SIIA), with specialised modules in Artificial Intelligence, Simulation, and Robotics.

My PhD, conducted at IMT Atlantique and ECAM Rennes - Louis de Broglie within the EASE team at IRISA, focused on **collective intelligence strategies for efficient autonomous industrial vehicles (AIVs) in Industry 4.0 contexts**: a shared-perception-based obstacle avoidance strategy, the adaptation of V2X communication standards to Industry 4.0, a decentralised dynamic task (re)allocation mechanism, a V2X data generation architecture, and fuzzy-model-based energy management strategies. Co-funded by Région Bretagne (ARED) and ECAM, the thesis received the **Best Thesis of the SPIN Doctoral School** award (2025). My work also received a Best Paper Award (ASPAI 2024), a Best Presentation Award (Colloque IMT 2024, ex aequo) and a Best Poster Award (Colloque IMT 2022, ex aequo). I hold the **CNU 27 qualification** (Computer Science).

Following my PhD, I worked for 3 months as a Research Engineer at IMT Atlantique on the **Projet Alpha**[1] (Alstef Group), applying my thesis ideas to airport baggage transport with AGV fleets (CARLA simulation, V2X cooperation).

Since February 2025, I am a Post-Doctoral Researcher at Naval Group, hosted in the IRL CROSSING, CNRS laboratory in Adelaide, Australia. My research focuses on **maritime simulation, multi-agent systems, and autonomous robotics in multi-domain environments** (aerial, surface, and underwater). A central part of my work is **LOTUSim**, an open-source real-time maritime simulation platform for human-vehicle teaming, and LOTUSim-Energy, its offshore-energy extension developed within the **FACET** project (French-Australian Government funding). Contributions include real-time physics-based multi-layer ocean current modelling, offshore wind wake modelling, and benchmarking of maritime simulators for swarm robotics and AI training, with 9 papers currently under review (3 Q1 journals, AAMAS 2027, 4 ICRA 2027, OMAE 2027) and 2 ANR proposals submitted (MariColab, EMERGE). I am a member of the **LOTUSim Technical Committee**, co-lead the **Maritime Working Group** at IRL CROSSING, served in 2026 as Naval Group representative on its Management Committee, and was elected non-permanent member of its **Laboratory Council** (August 2026).

In parallel, I have been involved in **supervision**: 1 PhD student in co-supervision (cotutelle IMT Atlantique / ANU, from 2026), 1 Research Engineer (2025–2026), 13 interns and 3 project groups across IRL CROSSING, Adelaide University, IMT Atlantique, and ECAM Rennes. I have served on several **hiring committees** at IRL CROSSING and ECAM Rennes, reviewed for IROS and AAMAS, and contributed to the **organisation of scientific events**, including the FACET Offshore Wind Maintenance Workshop (Adelaide, July 2025), PFIA 2022 and JJCR 2022.

International mobilities include a 3-month research visit to the Tsukada Laboratory, University of Tokyo (Japan, 2023), a 1-week visit to the Naval Group Center of Excellence in Singapore (August 2025), a technical internship at KMUTT Bangkok (Thailand, 2019), and an academic exchange at the Université de Sherbrooke (Canada, 2018). I have also been involved in **science outreach**: "Ma Thèse en 180 secondes" (4th prize, Rennes qualification round, 2024) and "L codent L créent" programming workshops for middle-school girls (8 sessions, Rennes, 2023).

[1] `https://alstefgroup.com/fr/alpha-project/`

Administration

Committee

2026 – present: Laboratory Council (IRL CROSSING) – Elected, Non-permanent
2026: Management committee (IRL CROSSING), Naval Group representative
2025 – present: Technical committee LOTUSim[2] (Naval Group)

Jury

Following my various experiences supervising students, I had the opportunity to participate in their internship defence jury's.

In addition, I also had the opportunity to serve on other evaluation panels:

2025 (Sep.): Jury member - 1 engineer internship defence, ENSTA.
2023 (Oct.): Jury member - 6 foreign-internship defences, ECAM Rennes apprentice engineers (in English).
2023 (Oct.): Jury member - 1 technical internship defence, ECAM Rennes.
2022 (Oct.): Jury member - 3 engineering-degree defences, ECAM Rennes.
2022 (Jan.): Jury member - 1 project defence, ECAM Rennes (3rd year students, Terrestrial Drones module).

[1] LOTUSim: open-source real-time maritime simulation platform for human-vehicle teaming developed by Naval Group. (`https://github.com/naval-group/LOTUSim`)

Teaching

During my PhD, I had the opportunity to supervise a total of 166 hours practical sessions and teach various audiences at ECAM Rennes - Louis de Broglie.

ECAM Rennes is a five-year engineering school that awards a degree equivalent to a Master's in General Engineering (student track), a Master's in Industrial Engineering (apprenticeship – GI), and a Master's in Digital Systems for Industrial Innovation (apprenticeship – SNII).

Courses and Responsibilities

ECAM Rennes - Louis de Broglie

Year	Theme	Vol.	Number	Type
2023–2024	ROS2	4h	1st year App. SNII	Lecture/TP
2023–2024	Distributed Systems	6h	3rd year Eng.	TP
2023–2024	Object-oriented programming in Python	12h	1st year Eng.	TP
2023–2024	Algorithmics with Python	18h	1st year Eng.	TP
2023–2024	Java	15h	1st year Eng.	TP
2022–2023	Terrestrial Drones Project	12h	3rd year Eng.	Project support
2022–2023	Distributed Systems	6h	3rd year Eng.	TP
2022–2023	Python	24h	1st year App. GI	TP
2022–2023	Java	18h	1st year Eng.	TP
2021–2022	Robotics Project	6h	2nd year Eng.	Project support
2021–2022	Python	24h	1st year App. GI	TP
2021–2022	Java	21h	1st year Eng.	TP

Responsibilities:

2023-2024: Distributed Systems - Creation of a 6-hours TP (2 sessions).

2023-2024: ROS2 - Creation of 1 hour introductory lecture and a 3 hours TP, intended for SNII apprenticeship students.

Course descriptions:

ROS2 (SNII apprentices, 1st year) – 1h lecture and 3h TP that I created to introduce ROS2: creating a package, a node, a topic and a service, using visualisation tools (RViz, rqt) and running a first simulation in Gazebo.

Distributed Systems (general engineering, 3rd year) – Introduction to distributed systems and concurrent programming: distributed algorithms, fault management, synchronous/asynchronous communication and key algorithms (broadcast, consensus, etc.). I created the 6h TP (2 sessions): a Python application simulating 4 autonomous vehicles at a central crossroads, in which students code a distributed algorithm so that the vehicles cross the intersection without collision.

Object-Oriented Programming in Python / Algorithmics with Python (general engineering, 1st year) – Review of the core concepts of imperative programming before functional/structured programming. Students learn to go from a problem statement to an algorithm design and its implementation in Python.

Java (general engineering, 1st year) – Fundamental concepts of object-oriented programming with Java: designing classes and objects to model complex problems and developing algorithmic skills with the Eclipse IDE. Lectures are complemented by teaching videos; all TPs are done in pairs and graded.

Terrestrial Drones Project (general engineering, 3rd year, IDI module) – Project-based module in which students learn project management, problem modelling, state-of-the-art review, design and development of software solutions, validation and professional documentation. I co-supervised the terrestrial drones project of 6 students.

Python (GI apprentices, 1st year) – Basics of programming in Python: variables, types, conditions, loops, functions and references. The last two TPs cover creating objects from classes and manipulating their data.

Robotics Project (general engineering, 2nd year) – Team project developing transversal, technical and interpersonal skills: students pool their ideas, time and skills towards a measurable performance compared with other teams. I followed one project team throughout the semester.

Student Supervision

PhD Co-supervision

2026 – 2029: Thomas Ung - Co-supervision (25 %)
Cotutelle: IMT Atlantique / ANU, IRL CROSSING.
Supervisors: C. Buche (IMT Atlantique), K. Daniell (ANU);
Co-supervisors: L. Bergantin, J. Grosset.
Topic: Internal Simulation for Adaptive Human–Robot Engagement.

Research Engineer Supervision

2025 – 2026: Marie Dubromel - Research Engineer, Naval Group, IRL CROSSING (Australia).
Topic: LOTUSim

Master Internships (7)

Host: IRL CROSSING, Adelaide (Australia)

2026: Loris Cazaux - Supervision (100 %)
Engineering apprentice (IT), ESIGELEC / RTE (France).
(5.5 months) Topic: Towards an Abstract and Modular Architecture for LOTUSim-Energy.

2026: Baptiste Piar - Supervision (50 %)
Master, Mines Paris – Paris Sciences et Lettres. (France).
(5 months) Topic: Building a Controller Baseline for AUVs - From PID Towards MPC under Flowing Currents.

Host: IMT Atlantique, Rennes (France)

2024: Juan Cuevas - Supervision (50 %)
MSc IoT Architecture and Engineering, IMT Atlantique.
(6 months) Topic: Real-world experimentation with robots of a cooperative intersection algorithm using ROS2 and V2X communication.

2024: Kazuto Matsumoto - Supervision (50 %)
Master of Computer Applications, University of Tokyo.
(2 months) Topic: Deep learning approaches for predicting obstacle position and velocity from virtual camera inputs in the CARLA simulator.

2024: Satoru Furuta - Supervision (50 %)
Master of Computer and Information Sciences, University of Tokyo.
(2 months) Topic: Camera-based obstacle detection in Gazebo and BIM-oriented environment reconstruction.

Host: ECAM Rennes – Louis de Broglie (France)

2024: Iheb Hamed - Supervision (50 %)
5th year engineering student, Polytech Tunis.
(5 months) Topic: Study and development of a multi-task mission management solution based on ground drones in a ROS2 environment.

2023: Rihab Binous - Supervision (50 %)
5th year engineering student, Polytech Tunis.
(5 months) Topic: Study and development of a multi-task mission management solution performed by heterogeneous mobile robots.

Bachelor Internships (6)

Host: IRL CROSSING, Adelaide (Australia)

2026: Ella James - Supervision (50 %)
Bachelor of Mathematical Sciences, Flinders University (Australia).
(3 months) Topic: Development and Comparison of Wind Models for Simulating Wind Turbine Energy Output.

2026: Riley Morrison - Supervision (50 %)
Bachelor of Computer Science, Flinders University (Australia).
(5 months) Topic: Migration of Vision-Based SLAM Pipeline to ROS2 for Lunar Rover Navigation, Validation and Deployment.

2026: Kartikeya Dashora - Supervision (50 %)
Bachelor of Engineering (Robotics), Flinders University (Australia).
(6 months) Topic: Real-Time MHE-based MPC for AUV Trajectory Tracking under Time-Varying Currents: simulation and Experimental Validation.

2026: Hayato Donoghue - Supervision (50 %)
Bachelor of IT (Game Development), Flinders University (Australia).
(3.5 months) Topic: LOTUSim Development and Enhancement through Game Design, Simulation Content Creation, and Environment Development.

2026: Bevan Shajan - Supervision (50 %)
Bachelor of Engineering (Robotics) (Honours), Flinders University (Australia).
(5 months) Topic: Integration of Crack Detection Algorithms into LOTUSim, Exploration of Event Cameras for Crack and Corrosion Detection.

2026: Gechhai Thornthea - Supervision (50 %)
Bachelor of Computer Science, Adelaide University (Australia).
(3 months) Topic: Integration of Gymnasium into LOTUSim.

Project-based Supervision

Adelaide University

2026: 5 students - Supervision (50%).
Topic: Integration of Pixhawk-based drone systems into LOTUSim using ROS2 and Gazebo, enabling manual control and mission-based simulation.

IMT ATlantique

2023–2024: 4 students - Supervision (50%).
Topic: Simulation of vehicle–infrastructure interactions using the CARLA Simulator (Unreal Engine), including the deployment of industrial robots in a simplified urban environment of Rennes and the study of a multi-task mission management mechanism for heterogeneous mobile robots.

ECAM Rennes - Louis de Broglie

2022–2023: 6 students - Supervision (50%).
Topic: Improvement of the performance and communication of a drone fleet.

Research

Key-words: Robotics, Distributed Systems, Artificial Intelligence, Simulation, Cooperative Autonomy, Maritime Context, ITS Context, Human-in-the-loop.

Publications

Book Chapter	*Inter. Journals*	*Inter. Conf.*	*National Conf.*	*Posters*	**Total**
1	6	8	3	4	**22**

Under review (9): 3 international journals (Q1), 4 ICRA 2027 (rank A*), 1 AAMAS 2027 (rank A), 1 OMAE 2027 (see Appendix, p. 21).
h-index : 5 (google scholar[3])
citations: 137 (google scholar)

Selected publications

Grosset J., Dubromel M., Lechêne H., Arzel Q. and Buche C. (2026). LOTUSim-Energy: A Maritime Simulator for Human-Drone Interaction in Autonomous Offshore Operation & Maintenance. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) – Workshop AQ²UASIM-v2*, Pittsburgh, USA, September 2026.

Buche C., **Grosset J.**, Lechêne H., Dubromel M., Havez-Bodivit P., Neo M. and Prodhon J. (2026). LOTUSim: Multi-Domain Simulator for Marine Robotics. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Pittsburgh, USA, September 2026. [Rank: A]

Grosset J., Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2024). Multi-Agent Simulation of Autonomous Industrial Vehicle Fleets: Towards Dynamic Task Allocation in V2X Cooperation Mode. *Integrated Computer-Aided Engineering*, 31(3):249–266. IOS Press. DOI: 10.3233/ICA-240735. [Q1]

Grosset J., Oukacha O., Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2024). Fuzzy Multi-Agent Simulation for Collective Energy Management of Autonomous Industrial Vehicle Fleets. *Algorithms*, 17(11):484. MDPI. DOI: 10.3390/a17110484. [Q2]

Grosset J., Ndao A., Fougères A.-J., Djoko-Kouam M., Couturier C. and Bonnin J.-M. (2023). A Cooperative Approach to Avoiding Obstacles and Collisions between Autonomous Industrial Vehicles in a Simulation Platform. *Integrated Computer-Aided Engineering*, 30(1):19–40. IOS Press. DOI: 10.3233/ICA-220694. [Q1]

Bouchabou D., **Grosset J.**, Nguyen S.M., Lohr C. and Puig X. (2023). A Smart Home Digital Twin to Support the Recognition of Activities of Daily Living. *Sensors*, 23(17):7586. MDPI. DOI: 10.3390/s23177586. [Q1]

[3] `https://scholar.google.com.au/citations?user=sf6Hqf0AAAAJ&hl=en`

Responsibility

Leadership

2026 – present: [Co-Leadership] Maritime working group, IRL CROSSING (Australia).
2025: [Leadership] LOTUSim open-source release (first open-source simulator in Naval Group's history).

Grants

"Cash": In addition to salaries, a budget allocated for my institution (not for the entire project).

Institutional Grants

Years	Project	Funding Source	Role	Topic	Cash	Position
26-27	FACET	French/Australian Gov	Member	Maritime simulation	135,000 €	Postdoc
25	FACET	French/Australian Gov	Member	Maritime simulation	50,000 AUD	–
					Partner: Reuniwatt	

Pending Grants

Project	Call	Coordinator	Role	ANR request
MariColab – human/multi-robot decision-authority sharing for maritime fleets (LOTUSim)	ANR PRC 2027	C. Buche	Member (WP4)	642 k€
EMERGE – renewable energy scheduling and energy-aware autonomous maintenance (LOTUSim-Energy)	ANR PRC 2027	G. Toogood	Member (WP4)	649 k€

Invited Speaker

2025: *SPIN Doctoral School Welcome Days* – PhD Prize ceremony (ED SPIN)[4]
2024: *Jeudi de la Recherche 2024*, ECAM Rennes – Louis de Broglie
2023: *Jeudi de la Recherche 2023*, ECAM Rennes – Louis de Broglie

International

Research Visits

2025 (1 week): Naval Group Center of Excellence, Singapore – Visit
2023 (3 months): University of Tokyo (Japan) – PhD Research Mobility visit, Tsukada Laboratory
2019 (3 months): KMUTT Bangkok (Thailand) – Internship in the Computer Science Laboratory

[3] https://ed-spin.doctorat-bretagne.fr/fr/actualites/prix-de-these-2025-decerne-juliette-grosset

Expertise

Reviewing

Journals: IEEE Sensors (2026), SN Computer Science (2026), Journal of Autonomous Intelligence (2023)

Conferences: IROS (2026), AAMAS (2026, 5 papers)

Hiring Positions

2026: Maëlys Cormont, Research Engineer Position - IRL CROSSING, CNRS (Australia).

2025: Marie Dubromel, Research Engineer Position - Naval Group, IRL CROSSING (Australia).

2023: Ouzna Oukacha, Post-Doctoral Position - ECAM Rennes - Louis de Broglie.

Organization of Scientific Events

2025: [Co-organizer] Workshop FACET[5] (Franco-Australian Indo-Pacific Centre for Energy Transition) - Offshore Wind Maintenance Workshop, Adelaide (Australia)

- Presentation of the market study on offshore wind maintenance commissioned by FACET to Naval Group (30 participants).
- Organization of a series of complementary activities focused on electrical networks, including site visits to research and industrial facilities.

2022: [Volunteer in the organizing committee] Plate-Forme Intelligence Artificielle (PFIA) 2022:

- 5-day event gathering researchers, industry professionals, and students
- 9 conferences, 5 workshops, 4 thematic days, and AI tutorials
- Presentation of a poster and a talk during the JFSMA young researchers session
- Participation in the AFIA workshop "Creating Collaborative, Inclusive and Fun Events in AI" - publication resulting from the workshop [13]

2022: [Member of the organization team] Journée des Jeunes Chercheurs et Jeunes Chercheuses en Robotique (JJCR 2022)

- Creation of the website[6]
- Management of submission system and paper uploads
- On-site support during the event (participant reception, session management, presentation coordination)

[5] https://www.franceaustraliaenergy.com/offshore-wind-maintenance-workshop---adelaide---july-2025

[6] https://jjcr-2022.sciencesconf.org/

Awards

Research Awards

2025: **Best Thesis Award** – SPIN Doctoral School (Brittany, France)
Awarded among approximately 160 PhD theses defended in 2024–2025, by a jury of 13 members.

2024: **Best Paper Award** – ASPAI 2024, 6th International Conference on Advances in Signal Processing and Artificial Intelligence, Funchal (Portugal).

2024: **Best Presentation Award** (ex aequo) – Colloque IMT 2024 "Responsible Industry of the Future", Aix-Marseille-Provence (France).

2022: **Best Poster Award** (ex aequo) – Colloque IMT 2022 "Information Sciences for New Mobilities", Palaiseau (France).

Science Outreach Awards

2024: **4th Jury Prize** – Ma Thèse en 180 secondes (MT180), Rennes qualifying round.

Scientific Groups Membership

2026 – present: Maritime Working Group, IRL CROSSING (Australia)
2022: GDR Robotique (JJCR 2022)
2022: French Artificial Intelligence Association (AFIA)

Science Communication

2024: Ma Thèse en 180 secondes
- 4th jury prize - Rennes qualification round
- Participation in the regional final in Brest
- Experience in science communication and popularization of research, adapting the presentation to diverse audiences

2023: L codent L créent[7] – 8 sessions of Python programming workshops for middle-school girls (Rennes).

Software

LOTUSim (Naval Group / IRL CROSSING)

LOTUSim is an open-source simulation ecosystem jointly developed within CROSSING to support research on Human–Autonomy Teaming. Initially initiated by Naval Group for maritime robotics, it has progressively evolved into a transversal scientific infrastructure shared by researchers from multiple institutions, research domains and Working Groups. LOTUSim has progressively become the **common experimental infrastructure supporting CROSSING research**. **LOTUSim-Energy**: real-time maritime simulation platform for human-drone teaming in offshore wind operations and maintenance (FACET project).

My role:
- Led the open-source release of LOTUSim (November 2025), the first simulator ever open-sourced by Naval Group.
- Member of the LOTUSim Technical Committee (2025–present).
- Development of LOTUSim and LOTUSim-Energy.

Related publications:
- Published: IROS 2026 Workshop [8] · IROS 2026 (rank A) [9]
- Under review:
  - ◇ Journal: Renewable Energy (Q1) [24]
  - ◇ Conferences 2027: AAMAS (rank A) [26] · ICRA (rank A*) [27, 28, 30] · OMAE [31]

Repositories:
- LOTUSim Documentation: irl-crossing-cnrs.github.io/LOTUSim
- LOTUSim: github.com/naval-group/LOTUSim
- LOTUSim-generic-scenario: github.com/IRL-Crossing-CNRS/LOTUSim-generic-scenario
- LOTUSimEnergy: github.com/IRL-Crossing-CNRS/LOTUSim-Energy

Simulators developed during my PhD

Autonomous Industrial Vehicles (AIV) Simulator

- Agent-based simulation of AIV fleets in Industry 4.0 settings: cooperative obstacle avoidance, dynamic task allocation and fuzzy energy management.
- **Related publications:** Algorithms 2024 (Q2) [4] · ICAE 2024 (Q1) [5] · ICAE 2023 (Q1) [6]
- **Repository:** gitlab.inria.fr/jgrosset/AIV_Simulator

CARLA–ROS2–OpenCDA Architecture

- Generation of V2X messages from the CARLA simulator for cooperative perception, developed during my research visit at the University of Tokyo.
- **Related publication:** VTC2024-Fall (rank B) [10]
- **Repository:** gitlab.inria.fr/jgrosset/carla-v2x-release

References

Pr. Cédric Buche
Professor, IMT Atlantique – Director of the CNRS IRL CROSSING, Adelaide (Australia)
Relationship: Post-doctoral supervisor (2025–present)
`cedric.buche@cnrs.fr`

Pr. Jean-Marie Bonnin
Professor, IMT Atlantique / IRISA, Rennes (France)
Relationship: PhD supervisor (2021–2024)
`jm.bonnin@imt-atlantique.fr`

Pr. Alain-Jérôme Fougères
Honorary Professor, ECAM Rennes – Louis de Broglie (France), former Head of the Computer Science Department
Relationship: PhD co-supervisor (2021–2024)
`alain-jerome.fougeres@ecam-rennes.fr`

Dr. Manabu Tsukada
Associate Professor, Graduate School of Information Science and Technology, The University of Tokyo (Japan) – Head of the Tsukada Laboratory
Relationship: Host of my PhD research visit (3 months, 2023)
`mtsukada@g.ecc.u-tokyo.ac.jp`

Dr. Thomas Chaffre
Lecturer, Flinders University, Adelaide (Australia)
Relationship: IRL CROSSING (2025–present)
`thomas.chaffre@flinders.edu.au`

Appendix

Supervision List

PhD Co-supervision (1)

2026 – 2029: Thomas Ung - Co-supervision (25 %), cotutelle IMT Atlantique / ANU
(36 months) Topic: Internal Simulation for adaptive human–robot engagement

Research Engineer Supervision (1)

2025 – 2026: Marie Dubromel - Research Engineer, Naval Group, IRL CROSSING
Topic: LOTUSim and LOTUSim-Energy

Internship Supervision – Master and Bachelor (13)

2026: Riley Morrison - Supervision (50 %)
(5 months) Topic: Migration of Vision-Based SLAM Pipeline to ROS2 for Lunar Rover Navigation, Validation and Deployment

2026: Kartikeya Dashora - Supervision (50 %)
(6 months) Topic: Real-Time MHE-based MPC for AUV Trajectory Tracking under Time-Varying Currents: simulation and Experimental Validation

2026: Hayato Donoghue - Supervision (50 %)
(3.5 months) Topic: LOTUSim Development and Enhancement through Game Design, Simulation Content Creation, and Environment Development

2026: Bevan Shajan - Supervision (50 %)
(5 months) Topic: Integration of Crack Detection Algorithms into LOTUSim, Exploration of Event Cameras for Crack and Corrosion Detection

2026: Gechhai Thornthea - Supervision (50 %)
(3 months) Topic: Integration of Gymnasium into LOTUSim

2026: Loris Cazaux - Supervision (100 %)
(5.5 months) Topic: Towards an Abstract and Modular Architecture for LOTUSim-Energy

2026: Baptiste Piar - Supervision (50 %)
(5 months) Topic: Building a Controller Baseline for AUVs - From PID Towards Real-Time MHE-based MPC under Time-Varying Currents

2026: Ella James - Supervision (50 %)
(3 months) Topic: Development and Comparison of Wind Models for Simulating Wind Turbine Energy Output

2024: Juan Cuevas - Supervision (50 %)
(6 months) Topic: Real-world experimentation with robots of a cooperative intersection algorithm using ROS2 and V2X communication.

2024: Kazuto Matsumoto - Supervision (50 %)
(2 months) Topic: Deep learning approaches for predicting obstacle position and velocity from virtual camera inputs in the CARLA simulator.

2024: Satoru Furuta - Supervision (50 %)
(2 months) Topic: Camera-based obstacle detection in Gazebo and BIM-oriented environment reconstruction.

2024: Iheb Hamed - Supervision (50 %)
(5 months) Topic: Study and development of a multi-task mission management solution based on ground drones in a ROS2 environment.

2023: Rihab Binous - Supervision (50 %)
(5 months) Topic: Study and development of a multi-task mission management solution performed by heterogeneous mobile robots.

Reviews List

Journals:

Name	Impact Fact.	Publisher	Rank	Year
IEEE Sensors	4.5	IEEE	Q1	2026
SN Computer Science	2.38	Springer Nature Link	Q2	2026
Journal of Autonomous Intelligence	1.05	Frontier Scientific Publishing		2023

Conferences:

Name	Rank	Year	Papers
IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)	A	2026	1
Int. Conf. on Autonomous Agents and Multiagent Systems (AAMAS)	A	2026	5

Grants List

Institutional Grants

2026 - 2027: [Member] LOTUSim Energy - FACET: French and Australian Government Funding
Topic: Maritime and Energy
≈ 135 K €

2025: [Member] LOTUSim Energy - FACET
Topic: Maritime and Energy
Collaboration with Reuniwatt.
≈ 50 K AUD

Pending Grants

2027 - 2031: [Member] MariColab - ANR AAPG 2027 (PRC)
Topic: Maritime collaborative autonomous and adaptive operations
Partners: CNRS (IRL CROSSING), Flinders University, Adelaide University, ENSTA, Naval Group.
≈ 642 K € requested (CNRS: ≈ 420 K €)

2027 - 2031: [Member] EMERGE - ANR AAPG 2027 (PRC)
Topic: Energy management ecosystem for renewables, grid and efficiency
Partners: CNRS (IRL CROSSING), Adelaide University, Flinders University, ENSTA, IMT Atlantique, ISAE-SUPAERO.
≈ 649 K € requested (CNRS: ≈ 410 K €)

Publications List

	Year	Book Chapter	Inter. Journal	Inter. Conf.	National Conf.	Poster
Post-Doc	2026			2		
	2025		2			
PhD	2024	1	2	3	1	
	2023		2		1	1
	2022			3	1	3
Total published		**1**	**6**	**8**	**3**	**4**
Under review			*3*	*6*		

Book Chapter (1)

[1] Grosset J., Fougères A.-J., Couturier C., Djoko-Kouam M. and Bonnin J.-M. Collision and Obstacle Avoidance for Industrial Autonomous Vehicles – Simulation and Experimentation Based on a Cooperative Approach. In *Advances in Robotics and Automatic Control*, vol. 3, pp. 1–23. IFSA Publishing, 2024. ISBN: 978-84-09-57872-6.

International Journal Articles (6)

[2] **Grosset J.**, Djoko-Kouam M., Fougères A.-J. and Bonnin J.-M. (2025). Fuzzy Agent-Based Simulations of Cooperative Strategies for Task Allocation, Collision Avoidance, and Battery Charging Management of Autonomous Industrial Vehicles *International Journal On Advances in Systems and Measurements*, 18(1&2), pp.8-18.

[3] **Grosset J.**, Fougères A.-J., Oukacha O., Djoko-Kouam M. and Bonnin J.-M. (2025). Simulation of Autonomous Industrial Vehicle Fleet Using Fuzzy Agents: Application to Task Allocation and Battery Charge Management *American Journal of Management*, 25(1). DOI: 10.33423/ajm.v25i1.7515 .

[4] **Grosset J.**, Oukacha O., Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2024). Fuzzy Multi-Agent Simulation for Collective Energy Management of Autonomous Industrial Vehicle Fleets. *Algorithms*, 17(11):484. MDPI. DOI: 10.3390/a17110484. [Rank: Q2]

[5] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2024). Multi-Agent Simulation of Autonomous Industrial Vehicle Fleets: Towards Dynamic Task Allocation in V2X Cooperation Mode. *Integrated Computer-Aided Engineering*, 31(3):249–266. IOS Press. DOI: 10.3233/ICA-240735. [Rank: Q1]

[6] **Grosset J.**, Ndao A., Fougères A.-J., Djoko-Kouam M., Couturier C. and Bonnin J.-M. (2023). A Cooperative Approach to Avoiding Obstacles and Collisions between Autonomous Industrial Vehicles in a Simulation Platform. *Integrated Computer-Aided Engineering*, 30(1):19–40. IOS Press. DOI: 10.3233/ICA-220694. [Rank: Q1]

[7] Bouchabou D., **Grosset J.**, Nguyen S.M., Lohr C. and Puig X. (2023). A Smart Home Digital Twin to Support the Recognition of Activities of Daily Living. *Sensors*, 23(17):7586. MDPI. DOI: 10.3390/s23177586. [Rank: Q1]

International Conference Articles (8)

[8] **Grosset J.**, Dubromel M., Lechêne H., Arzel Q. and Buche C. (2026). LOTUSim-Energy: A Maritime Simulator for Human-Drone Interaction in Autonomous Offshore Operation & Maintenance. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) – Workshop AQ²UASIM-v2*, Pittsburgh, USA, September 2026.

[9] Buche C., **Grosset J.**, Lechêne H., Dubromel M., Havez-Bodivit P., Neo M. and Prodhon J. (2026). LOTUSim: Multi-Domain Simulator for Marine Robotics. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Pittsburgh, USA, September 2026. [Rank: A]

[10] **Grosset J.**, Bonnin J.-M., Fougères A.-J., Tsukada M. and Djoko-Kouam M. (2024). Generation of V2X Messages from Carla Simulator for Cooperative Perception: Application to Pedestrian Safety. *IEEE 100th Vehicular Technology Conference (VTC2024-Fall)*, Washington D.C., USA, October 2024. [Rank: B]

[11] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2024). Fuzzy Agent-Based Simulation of Integrated Solutions for Task Allocation and Battery Charge Management for Fleets of Autonomous Industrial Vehicles. *AI-based Systems and Services (AISyS) 2024*, Venice, Italy, October 2024.

[12] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2024). Fuzzy Agent-Based Simulation for Managing Battery Recharging for a Fleet of Autonomous Industrial Vehicles. *ASPAI 2024: 6th International Conference on Advances in Signal Processing and Artificial Intelligence*, Funchal (Madeira), Portugal, April 2024. **Best Paper Award.**

[13] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2022). Collective Obstacle Avoidance Strategy – an Agent-Based Simulation Approach. *ASPAI 2022: 4th International Conference on Advances in Signal Processing and Artificial Intelligence*, Corfu, Greece, October 2022.

[14] **Grosset J.**, Fougères A.-J., Djoko-Kouam M., Couturier C. and Bonnin J.-M. (2022). Simulation of a Collision and Obstacle Avoidance Algorithm for Cooperative Industrial Autonomous Vehicles. *2nd IFSA Winter Conference on Automation, Robotics & Communications for Industry 4.0 (ARCI' 2022)*, Andorra la Vella, Andorra, February 2022.

[15] Bosser A.-G., Ermakova L., de Saint-Cyr F.D., De Loor P., Charpenay V., Pépin-Hermann N., Alcaraz B., Autran J.-V., Devillers A. and **Grosset J.** (2022). Poetic or Humorous Text Generation: Jam Event at PFIA2022. *13th Conference and Labs of the Evaluation Forum (CLEF 2022)*, vol. 3180, pp. 1719–1726. CEUR-WS.org.

National Communications (3)

[16] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2024). Multi-Agent Simulation of Autonomous Industrial Vehicle Fleets. Collective Energy Management with V2X Cooperation. *Colloque IMT 2024: Industrie du Futur Responsable*, Mines Saint-Étienne – Campus Aix-Marseille-Provence, April 2024. **Best Presentation Award ex-aequo.**

[17] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2023). Using AI for Cooperative Perception. *Colloque IMT 2023: Data & AI*, Nantes, November 2023.

[18] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2022). Collective Intelligence Strategies for Efficient Autonomous Industrial Vehicles. From simulation to real experiments. *JFSMA 2022 – Sessions jeunes chercheurs*, École des Mines, Saint-Étienne, July 2022.

Posters (4)

[19] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2023). Collective Intelligence Strategies for Efficient Autonomous Industrial Vehicles. Traffic hazards simulated by fuzzy agents. *32ème Rencontres Francophones sur la Logique Floue et ses Applications (LFA 2023)*, Bourges, November 2023.

[20] **Grosset J.**, Fougères A.-J., Djoko-Kouam M., Couturier C. and Bonnin J.-M. (2022). Collective Intelligence Strategies for Efficient Autonomous Industrial Vehicles. From simulation to real experiments. *Colloque IMT 2022: Les sciences de l'information au service des nouvelles mobilités*, Palaiseau, October 2022. **Best Poster Award ex-aequo.**

[21] **Grosset J.**, Fougères A.-J., Djoko-Kouam M. and Bonnin J.-M. (2022). Collective Intelligence Strategies for Efficient Autonomous Industrial Vehicles. From simulation to real experiments. *JFSMA 2022 – Sessions jeunes chercheurs*, École des Mines, Saint-Étienne, July 2022.

[22] **Grosset J.**, Fougères A.-J., Djoko-Kouam M., Couturier C. and Bonnin J.-M. (2022). Towards increasing the autonomy of Autonomous Industrial Vehicles using collective intelligence strategies. *Jeudi de la Recherche 2022*, ECAM Louis de Broglie, March 2022.

Publications Under Review

International Journal Articles (3)

[23] **Grosset J.**, Fougères A.-J. and Bonnin J.-M. (2026). Cooperative Autonomy Scale for Autonomous Industrial Vehicles Fleets. *Transportation Engineering*, Elsevier. Under review. [Rank: Q1]

[24] Dubromel M., **Grosset J.**, James E., Cazaux L. and Buche C. (2026). Benchmarking Engineering Wake Models for Real-Time Offshore Wind Farm Simulation: Power and Spatial-Field Accuracy Against Actuator-Line CFD and FLORIS. *Renewable Energy*, Elsevier. Under review. [Rank: Q1]

[25] **Grosset J.**, Dubromel M. and Buche C. (2026). Do Existing Maritime Simulators Support Large Robot Swarms? A Real-Time and Accelerated-Time Perspective. *Robotics and Autonomous Systems*, Elsevier. Under review. [Rank: Q1]

International Conference Articles (6)

[26] **Grosset J.**, Dubromel M. and Buche C. (2027). How Many Robots Can Your Simulator Handle? A Multi-Agent Scalability Benchmark of Open-Source Maritime Simulators. *International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, May 2027. Under review. [Rank: A]

[27] **Grosset J.**, Dubromel M. and Buche C. (2027). Which Maritime Simulator for Swarm Robotics? A Unified Benchmark of Open-Source Platforms for Human Interaction and AI Training. *IEEE International Conference on Robotics and Automation (ICRA 2027)*. Under review. [Rank: A*]

[28] **Grosset J.**, Dubromel M. and Buche C. (2027). A Physically Consistent Current Model for Marine Robotics Simulation: From Ekman Layers to Closed-Loop Control. *IEEE International Conference on Robotics and Automation (ICRA 2027)*. Under review. [Rank: A*]

[29] Piar B., **Grosset J.**, Chaffre T., Dubromel M., Dashora K., Wheare J., Lammas A., Sammut K. and Buche C. (2027). A Controlled Ablation of the Disturbance Observer in Nonlinear MPC for AUVs under Time-Varying Currents. *IEEE International Conference on Robotics and Automation (ICRA 2027)*. Under review. [Rank: A*]

[30] Dubromel M., **Grosset J.**, James E., Cazaux L., Arzel Q. and Buche C. (2027). What a Turbine Wake Does to an Inspection UAV: Wake Simulation for Offshore Wind Autonomous Operation & Maintenance. *IEEE International Conference on Robotics and Automation (ICRA 2027)*. Under review. [Rank: A*]

[31] **Grosset J.**, Dubromel M., Arzel Q., Lechêne H. and Buche C. (2027). From Market Barriers to Simulation Requirements: A Gap Analysis of Tools for Autonomous Offshore Wind O&M in Australia. *ASME International Conference on Ocean, Offshore and Arctic Engineering (OMAE 2027)*. Under review.